Teaching Notes 9B Wrap Up*

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*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

What Were We Hoping to Accomplish?

Two Parts to the Course…

Part I: Capital Budgeting

- What real assets should the firm invest in?
- Project valuation
- PV, discount rates, valuation in practice (WACC, APV)

1 **Valuation Basics**

- **Free Cash Flows, Discount Rates, and NPV**

2 **Direct Applications**

- **Cost of Capital**
- **WACC and APV**

3 **Asset Pricing/Investment Fundamentals**

Two Parts to the Course…

Part II: Financial Policy

- How should projects be financed?
- Debt, equity, information problems, taxes
- Payout policy

1 **Financial Policy Basics**

- **The "five" assumptions of Modigliani-Miller**

2 **Debt or Equity?**

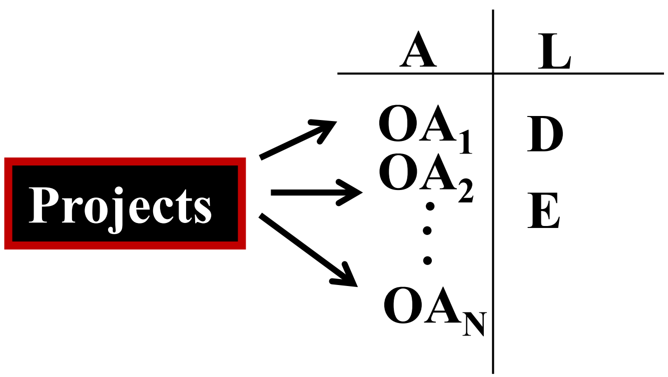
- **Wealth Transfer/Debt Overhang/Risk Shifting**

3 **Dividends or Cash?**

- **Signaling/Catering/Buffer**

Capital Budgeting

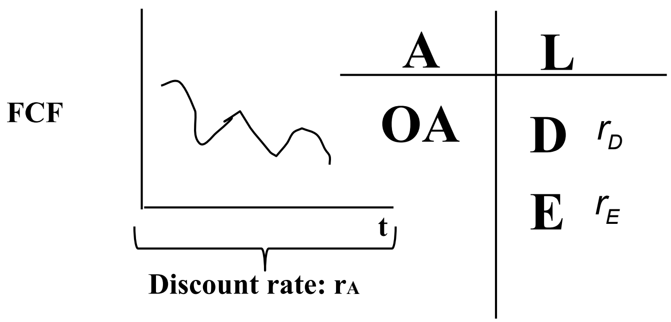
Present Value & Discounted Cash Flows



- A Firm is a collection of projects
  - Each project is producing (or consuming) cash flows at different points in time
  - Our job is to figure out a way to compare different projects so that managers can pick the one that is appropriate
  - Use Discounted Cash Flows to do this
- Present Value formula allows us to move cash flows across time and compute a value at time 0
  - $PV = C_1/(1+r) + C_2/(1+r)^2 + \ldots + C_T/(1+r)^T$
  - Recall Special Cases:
    - Standard Perpetuity/Growing Perpetuity
    - Annuity/Growing Annuity
  - Compute NPV of the project based on PV of all benefits and costs to determine which project to take

Lecture 1A + Lecture 0

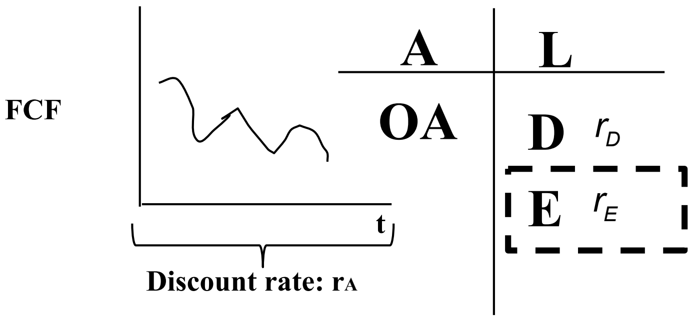
Value of a Firm: The Fundamentals



- We discussed that computing NPV requires we know two things:
  - Cash Flows: Numerator of PV formula
  - Discount Rates: Denominator of PV formula
- FCF in any given period t is
  - $\mathbf{FCF_t} = EBIT_t(1 - t) + Dep_t - CAPX_t - \Delta NWC_t$
  - The challenge lies in extracting the above information from the income statement and the balance sheet
  - Notes:
    - In this formula, ignore interest expense (<u>even if present</u>) in the income statement
    - Remember to account for Terminal Value using one of the two methods discussed
    - Focus only on "incremental" cash flows

Stock Valuation

- If we focus on the liability side, one of the stakeholders is equity holders (E)
  - Their expected returns r_E reflect returns promised to them
  - The stock price reflects the potential cash flow stream they expect to get in the future through dividend payments



$$P_0 = \sum_{t=1}^{\infty} \frac{DIV_t}{(1+r)^t}$$

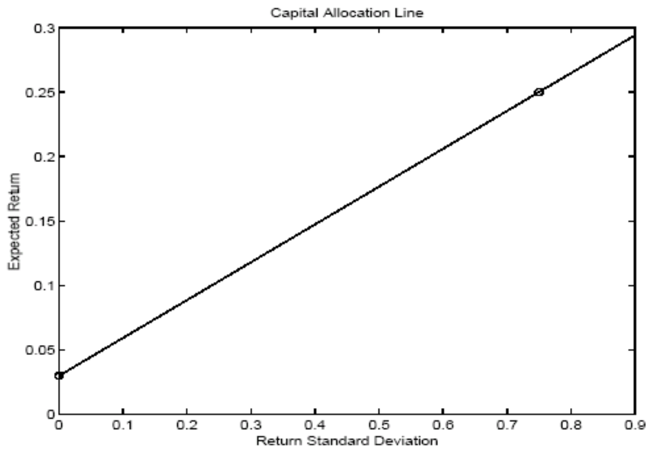
$$P_0 = \frac{DIV_1}{r-g}$$

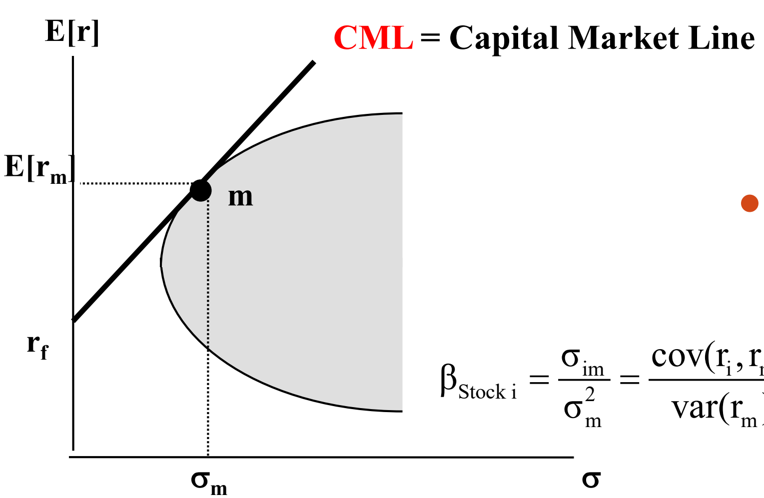
$$P_0 = \frac{EPS_1}{r} + PV(GO)$$

  - Under certain assumptions, this formula collapses into a growing perpetuity-like formula
  - With more assumptions, it can be rewritten as a function of current earnings and the PV of future growth opportunities (GO)

Portfolio Choice & Diversification

$$E[r_p] = r_f + \frac{\sigma_p}{\sigma_A} \times (E[r_A] - r_f) = r_f + \frac{[E[r_A] - r_f]}{\sigma_A} \times \sigma_p$$





$$\beta_{\text{Stock i}} = \frac{\sigma_{im}}{\sigma_m^2} = \frac{\text{cov}(r_i, r_m)}{\text{var}(r_m)}$$

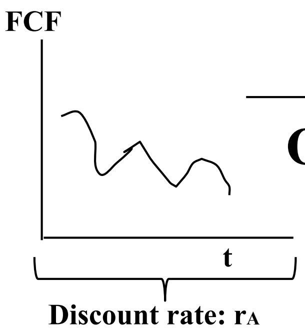
- Investment evaluated based on two metrics – its "risk" and its "expected return"
- <u>Case 1: One risky asset & risk free asset</u>
  - Results in Capital Allocation Line (CAL)
- <u>Case 2: Two risky assets</u>
  - Diversification – due to covariance between the two asset returns, risk of the portfolio went down
  - However, as we combine many assets, part of the risk cannot be diversified away – systematic risk
  - A frontier helps us determine how to invest in these assets – called the mean variance frontier
  - The lower part of the frontier is "inefficient"
- <u>Case 3: Many risky assets and risk free asset</u>
  - Allocation based on mix between risk free asset & the market portfolio on the efficient frontier
  - Contribution of an individual stock to overall risk of the market portfolio given by its beta
  - Similar formula as Case 1 applies

$$E(r_p) = r_f + \frac{\sigma_p}{\sigma_m} \times (E(r_m) - r_f) = r_f + \frac{[E(r_m) - r_f]}{\sigma_m} \times \sigma_p$$

CAPM

$$r_i = r_f + \beta_i * (r_m - r_f)$$

$$\beta_i = \frac{cov(r_i, r_m)}{var(r_m)} = \frac{\sigma_{i,m}}{\sigma_m^2}$$



$$r_A = r_f + \beta_A * (r_m - r_f)$$

A	L
OA	D r_D $r_D = r_f + \beta_D * (r_m - r_f)$
	E r_E $r_E = r_f + \beta_E * (r_m - r_f)$

- The expected return on any given security is a function of three variables: the risk free rate (r_f), the beta of the security with respect to the market portfolio (β_i), and the market risk premium ($r_m - r_f$)
  - Only the part of the security that contributes to systematic risk is compensated. The other part called idiosyncratic risk is diversified away.
- β_i can be interpreted as the regression coefficient in the regression

$$\tilde{r}_{i,t} - r_{f,t} = \alpha_i + \beta_i \times \left(\tilde{r}_{m,t} - r_{f,t}\right) + \tilde{\varepsilon}_{i,t}$$

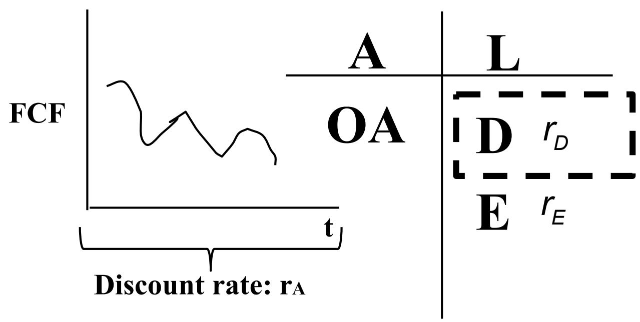
$$\beta_i^2 \times \sigma_m^2$$

$$\sigma_\varepsilon^2$$

  - Systematic risk in this framework
  - Idiosyncratic risk in this framework
  - The regression R^2 measures the fraction of total firm *i* risk that is due to market risk (i.e. the fraction of risk that gets compensated)

$$R^2 = \frac{\text{Systematic Risk}}{\text{Total Risk}} = \frac{\beta_i^2 \times \sigma_m^2}{\sigma_i^2} = \frac{\beta_i^2 \times \sigma_m^2}{\beta_i^2 \times \sigma_m^2 + \sigma_\varepsilon^2}$$

Bond Valuation



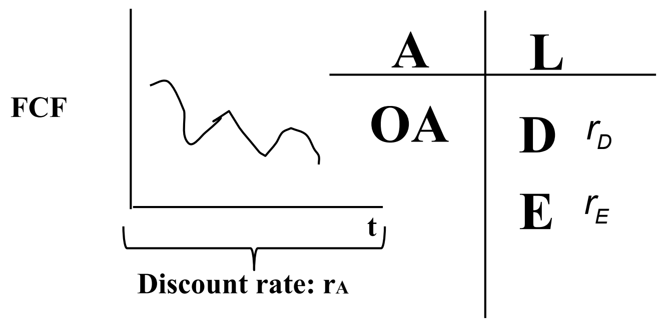
- If we focus on the liability side, one of the stakeholders is debt holders (D)
  - Their expected returns r_D reflect returns promised to them
  - The “bond” price reflects the potential cash flow payments they expect to get in the future in the form of periodic coupon payments and end of period face payment

$$B(C,T) = \sum_{t=1}^{T} \frac{C_t}{(1+r)^t} + \frac{F}{(1+r)^T}$$

$$B(0,T) = \frac{F}{(1+r)^T}$$

  - Sometimes, a zero coupon bond (i.e., a bond that only pays at the end of the period) is useful
  - Using zero coupon bonds of the US government, we can construct a yield curve. This lets us infer the risk free rate used by investors for lending money over a fixed horizon in the future.

NPV based on Lectures so far…



Find firms with similar business risk

↓

Keep those with "usable" β_E

↓

Apply the "magic"/unlever formula

↓

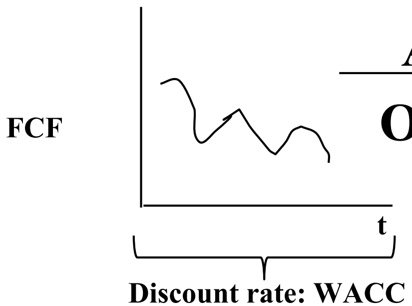
Average β_A across comps = Project β_A

↓

Apply CAPM

- FCF every period
  - **FCF$_t$** = $EBIT_t(1 - t) + Dep_t - CAPX_t - \Delta NWC_t$
  - Notes:
    - In this calculation, we ignore interest expense (even if present) in the income statement
    - Be careful with the timing of cash flows when discounting terminal value using perpetuity method
- Project discount rate r_A
  - β_A and r_A are two sides of a coin
  - Capture "business risk"
  - Use comparable or twin method to derive β_A and r_A
    - 5 step recipe
  - Note:
    - Capital structure of the project is irrelevant for β_A or r_A

Total Value is Higher with Debt due to DTS: Account for it using WACC



A	L	
OA	[D-DTS]	$r_D(1-t_C)$
	E	r_E

- FCF same as before
  - Ignore interest expense (even if present) when doing calculations
- Get discount rate, which will be "adjusted" to account for higher value due to DTS
  - Called WACC

$$WACC = r_E \frac{E}{E+D} + r_D(1-t_C)\frac{D}{E+D}$$

- Note: Be careful when discounting terminal value
  - Usually assume growing perpetuity
  - Discounting done by WACC

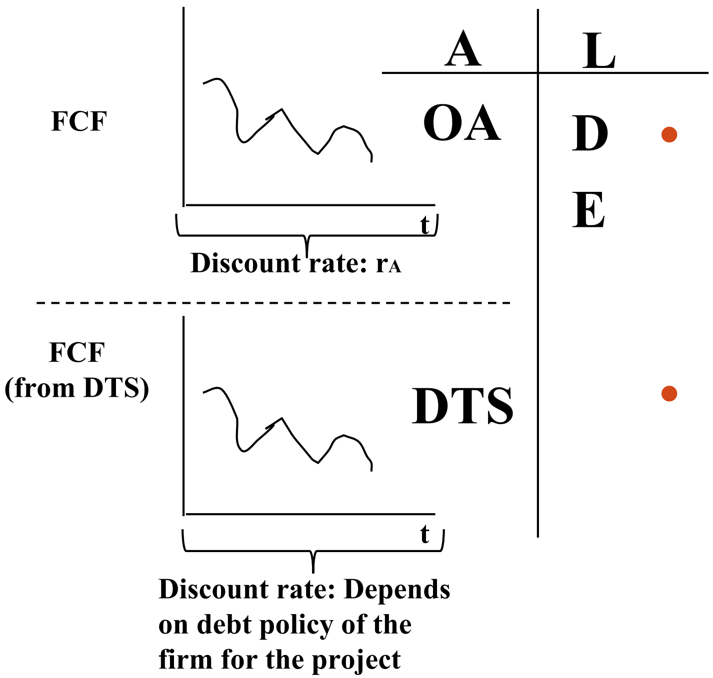
Ameritrade Case

- Estimated the cost of capital
  - Used "Twin Method" to measure risk
  - Calculated equity betas for comparables
  - Unlevered equity betas to get asset betas
  - Finally, the cost of capital came out of CAPM

- $r_A = r_f + \beta_A (r_m - r_f)$
  - r_f ?
  - $(r_m - r_f)$?

- β_A
  - **How pick "comps"?**
  - **How many to pick?**
  - **Over what time horizon to compute β_E of comps?**
  - **What should we average?**

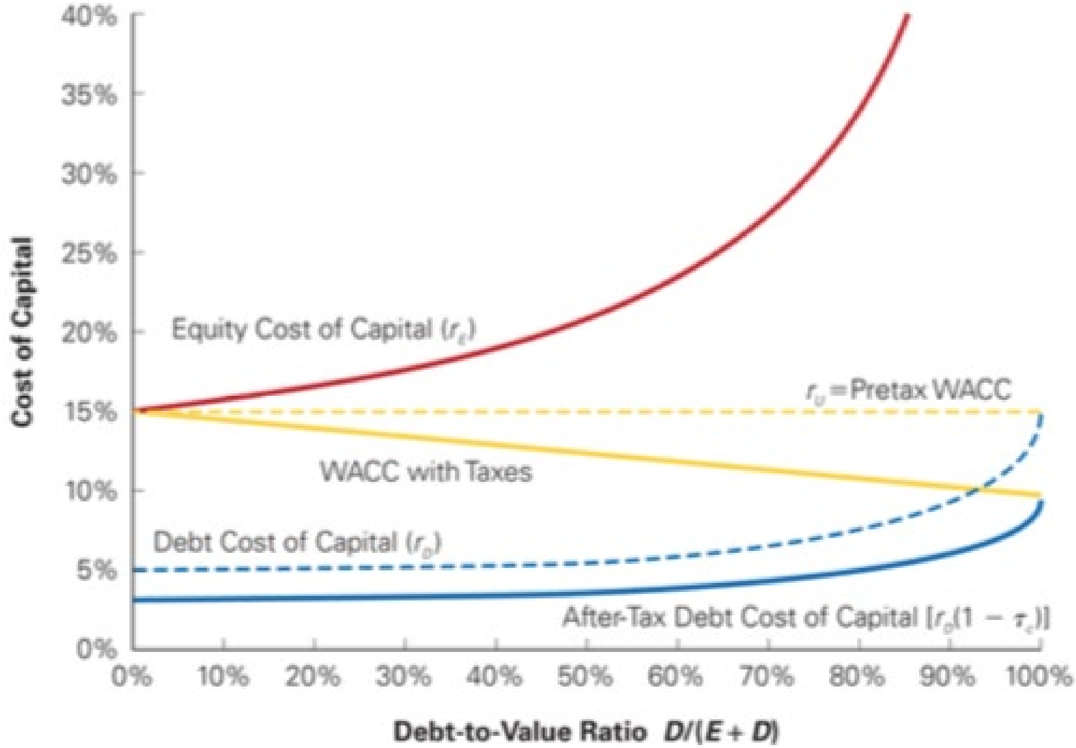
Lecture 5B Total Value is Higher with Debt due to DTS: Account for it using APV



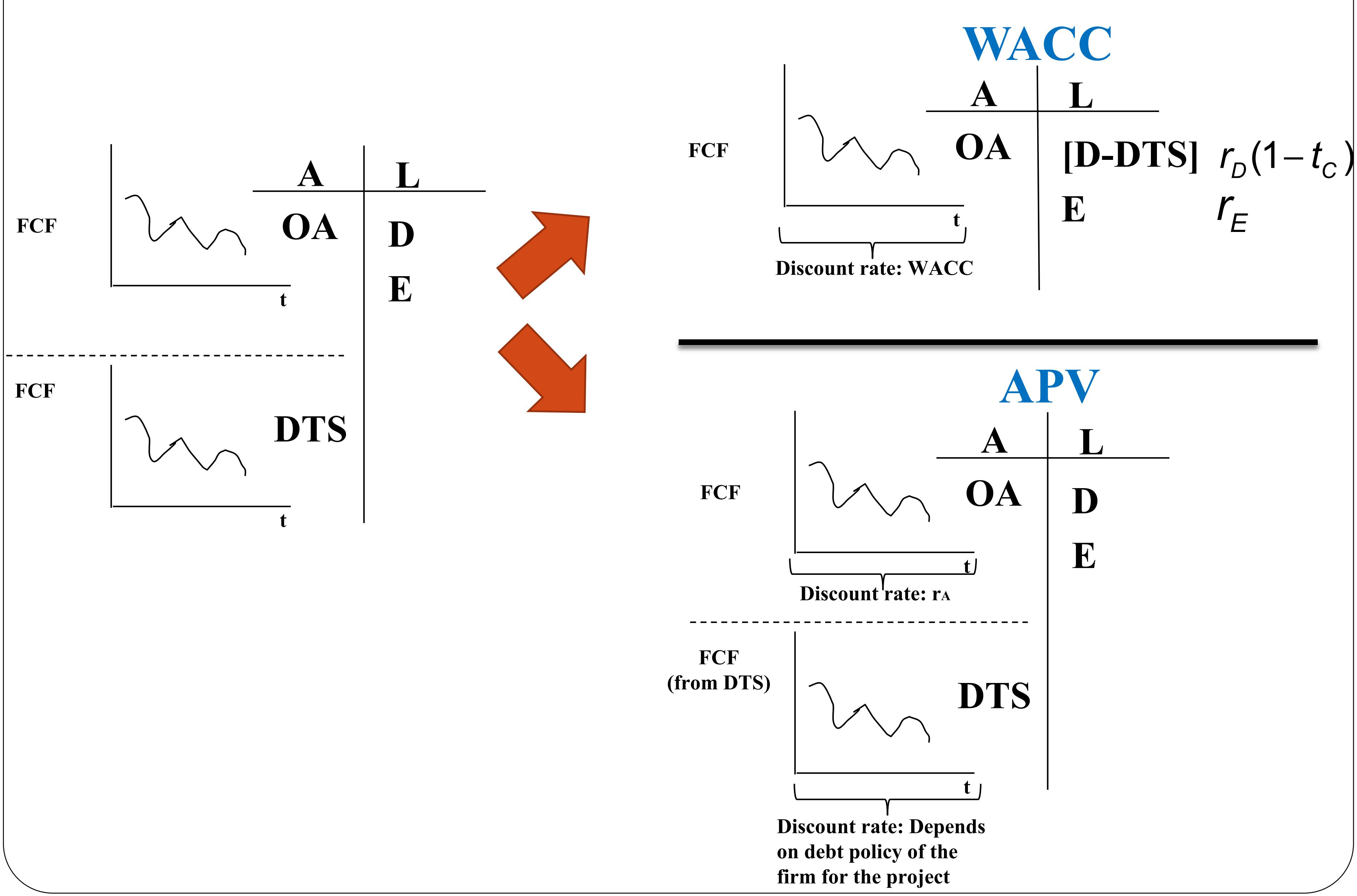
- $APV = NPV_{AE} + PV(DTS)$
  - Value FCFs from two projects on the asset side: (i) OA and (ii) DTS
- FCFs from OA project: NPV_{AE}
  - FCF same as before
    - Ignore interest expense when doing calculations
  - Discount using r_A
- FCFs from DTS project: PV(DTS)
  - FCF from DTS in any time period t
    - Tax rate * Interest_t = Tax rate * D_t * r_D
  - For discounting DTS, the discount rate for it depends on debt policy of the firm for the project
    - r_D if Debt fixed over project horizon
    - r_A if Debt/Value (Debt/Equity) is fixed over project horizon
- Note: Be careful with terminal value
  - Separately calculate terminal value of OA project
    - Discount rate for OA project is r_A
  - Separately calculate terminal value of DTS project
    - Discount rate for DTS project depends on debt policy

Snap Case

- Understand "all-equity" value of the project
  - FCFs
    - What is the terminal value?
- APV vs. WACC?
  - Both use FCFs from above
    - Differences in terminal value calculations
  - APV adds PV(DTS) while WACC adjusts the discount rate to account for financing effects



WACC and APV: Big Picture



Real World: Facts

$$V = \sum_t \frac{CF_t}{(1+r)^t}$$

- V = Value of Assets in Place + Value of Growth Opportunities
- Bulk of the value is in "g"

Uber Revenue Slows as Quarterly Loss Surges to $1.1 Billion

By Eric Newcomer
November 14, 2018, 3:00 PM CST

- Uber generated $2.95 billion in revenue in the third quarter
- Losses widened 20 percent compared with the previous quarter

Uber Proposals Value Company at $120 Billion in a Possible IPO

Eye-popping offering, which could take place early next year, is nearly double the ride-hailing company's valuation in a fundraising round two months ago

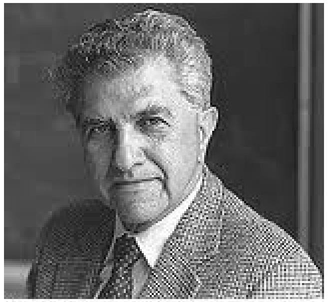
By Liz Hoffman, Greg Bensinger and Maureen Farrell
Updated Oct. 16, 2018 1:28 p.m. ET

Financial Policy

Financial Policy Basics

- Defining the value of a corporation
  - Value of corporate assets must equal $E_{MV} + D_{MV}$

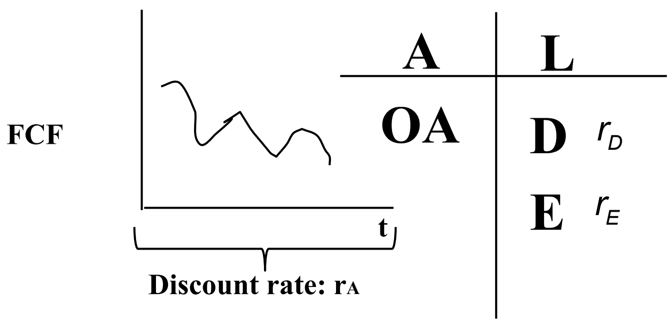
Miller



Modigliani

- Modigliani-Miller
  - When evaluating financial policy, start by asking: “What would MM say?”
    - MM usually says it doesn’t matter
    - If it matters, it is because an MM assumption fails…which one?
  - Note: MM World does not mean there is no risk
    - Discount rate is r_A

Value of Firm is Independent of whether D or E in Capital Structure (MM)



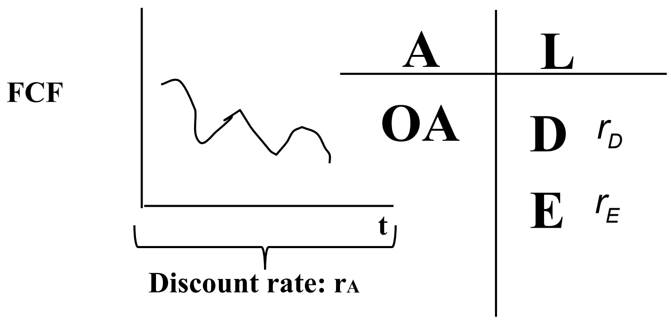
- MM Proposition says that value of the firm does not depend on whether you choose Debt or Equity to finance the project
- Value of the firm is entirely driven by its operating asset side
  - Value of OA project based on FCF and discount rate
  - FCF given as per Lecture 1
  - Discount rate given as r_A
- MM proposition holds when "five assumptions" are true
  - Gives a useful baseline and provides a framework through which all financial decisions can be evaluated

The "Five" MM Assumptions

- MM holds under these assumptions:
  - Perfect and complete capital markets
  - No taxes
  - Bankruptcy is not costly
  - Capital structure does not affect investment policy and cash-flows
  - Symmetric information

Relaxing Assumption #1:

Markets are Imperfect and Incomplete



- **Imperfect capital markets**
  - ➢ **Leads to profitable arbitrage opportunities**
  - ➢ **Not something that is likely to influence marginal decision to issue Debt versus Equity for a long term horizon project**

- **Incomplete capital markets**
  - ➢ **Leads to Clientele effect**
  - ➢ **Not something that firms want to exploit by tilting their capital structure. Rather, let the investment banks with comparative advantage do it at an efficient scale**

- **Both lead to interesting phenomena but do not think this matters for long term capital structure decisions of managers**

Relaxing Assumption #1:

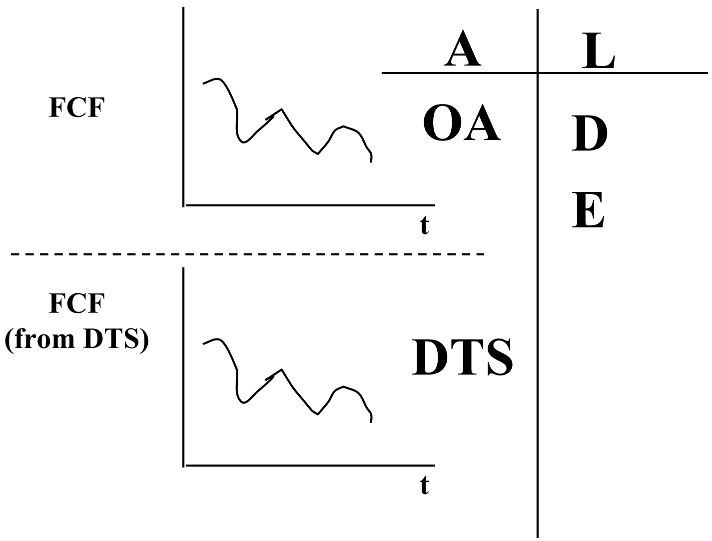
Markets are Imperfect and Incomplete

- Perfect capital markets argument can fail in two ways
  - Market prices securities "incorrectly" (results in "anomalies")
  - Investors cannot trade "as they wish" (results in "clienteles")

- <u>In the real world…</u>
  - Market efficiency anomalies exist
    - Small stock premium, value premium, momentum, etc.
    - Harvey, Liu, and Zhu (2016) consider 316 documented "anomalies"
  - Clienteles exist
    - Eg: Pension funds in Marriott
  - But do not think either of these matter for long-term capital structure decisions

Relaxing Assumption #2:

There are Taxes

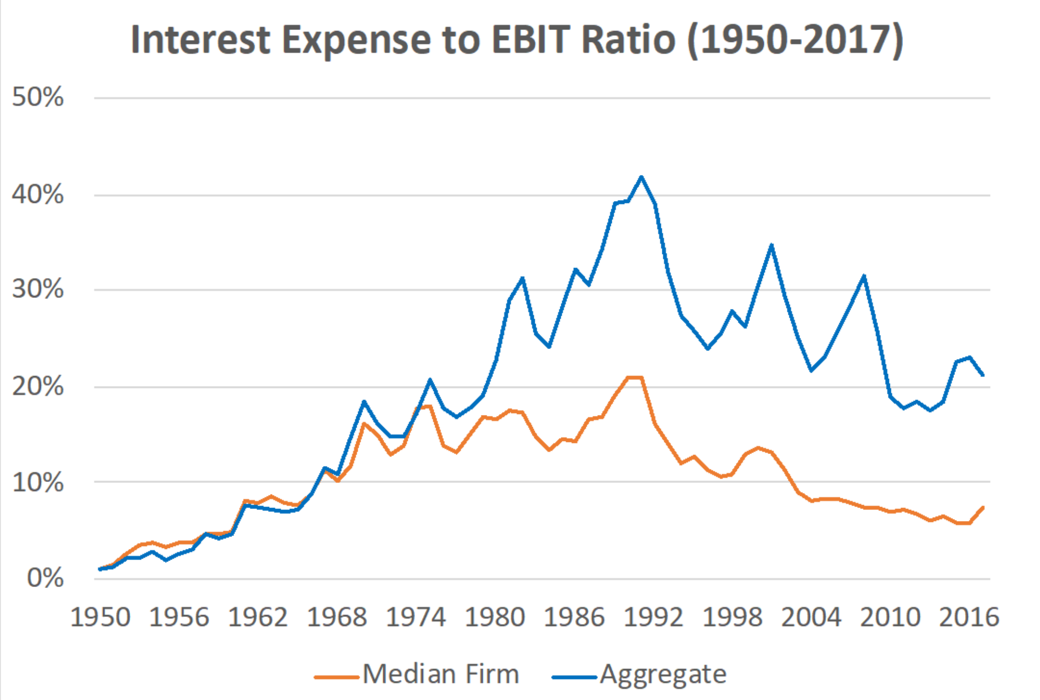


- Debt brings with it a debt tax shield
  - ➢ With perpetual debt, we have been using the formula PV(DTS) = D x t_c
  - ➢ But other taxes exist – like individual taxes on interest payments and individual taxes on dividends
  - ➢ Together, they can change the formula to
    - ➢ PV(DTS) = D x t*, where $t^*=[1-(1-t_c)(1-t_e)/(1-t_p)]$
    - ➢ t_p and t_e are individual taxes on interest income and dividends
  - ➢ So why ignore this formula?
    - ➢ Not because other taxes are small.
    - ➢ But because the marginal investor who invests in capital structure of a firm is an institution (like a pension fund) that has very low t_p and t_e. When these taxes are small, the messy formula collapses to D x t_c
- The benefit of the DTS goes to equity holders
- If taxes were the only friction that impacted capital structure, managers would issue Debt to the point that they extinguished EBIT period by period
  - ➢ D/V for average firm out there would be 50-60%

Relaxing Assumption #2

- DTS in a year = Interest Payment * Corporate Taxes
  - Special case (Perpetuity discounted at r_D): $PV(DTS) = \frac{\tau_C r_D D}{r_D} = \tau_C D$
  - Personal taxes reduce the value of the DTS: $PV(DTS) = \left[1 - \frac{(1-\tau_C)(1-\tau_e)}{(1-\tau_p)}\right] D$
  - Marginal investor is a large institution $\Rightarrow \tau_e = \tau_p = 0$. The general formula collapses to $PV(DTS) = \tau_C D$
  - If taxes were the only consideration, firms would increase leverage to maximize the DTS (ie: take debt such that EBIT = Interest Payments)

- <u>In the real world…</u>

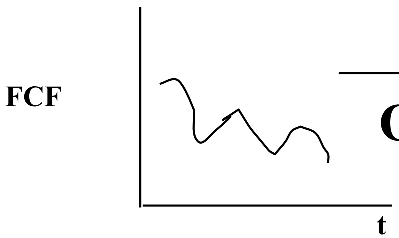


Data: Compustat

Lecture 7B

Relaxing Assumption #3:

Bankruptcy is Costly



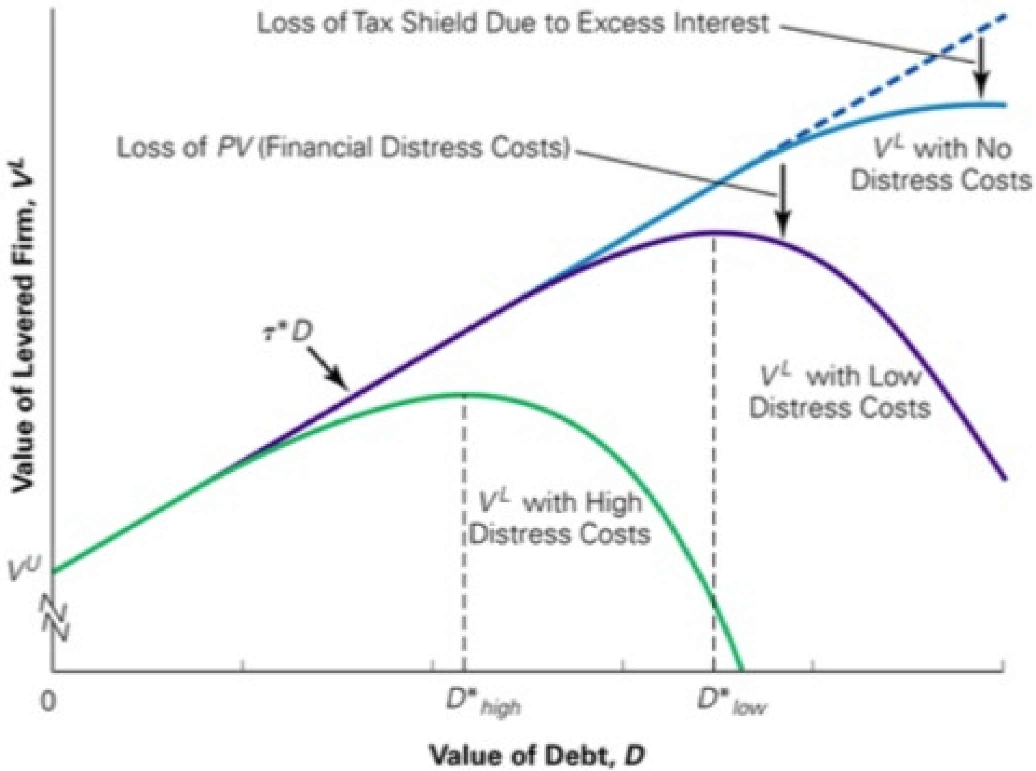
A	L
OA	D
	E
+ PV(DTS)	
- PV(CFD)	

- Debt also brings costs of financial distress (CFD)
  - Economic distress impacts all firms in a given industry equally (bad shocks)
  - Financial distress is NOT economic distress. It is amplification of economic shocks due to presence of debt
- Two types of CFD
  - Direct costs: Incurred after firm goes into bankruptcy (eg: legal fees). When quantified, are not large relative to PV(DTS)
  - Indirect costs: Incurred in "anticipation" of firm facing financial distress in the future (eg: customers leaving). When quantified, are large relative to PV(DTS)
- CFD also borne by the equity holder
- Managers "trade-off" the benefit that debt brings (PV(DTS)) with the costs that debt brings (PV(CFD)).
- Trade-off model explains most patterns of capital structure used by firms in the real world.
  - For each factor, ask: when the factor goes up, does it change CFD or DTS? Then see if that implies an increase or decrease in Debt as per Trade-off theory

Relaxing Assumption #3:

Bankruptcy is Costly

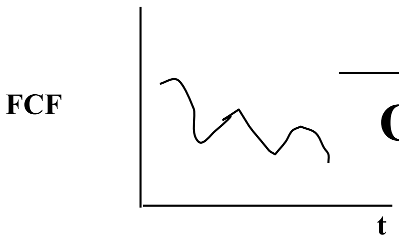
- We made bankruptcy costly.
  - Cost of financial distress (not economic distress) matters
- <u>In real world…</u>
  - Direct costs: Small
  - Indirect costs (losses from anticipated future distress): Large



- Cost of Financial Distress (CFD) borne by shareholders ex-ante (not by debt-holders)
- The benefit of DTS and the cost of CFD lead to the Trade-Off theory of capital structure
  - "Optimal" D/E ratio balances benefit of DTS vs. CFD
    - Explains patterns in leverage
    - Does not explain why profitable firms have less leverage

Relaxing Assumption #4:

Capital Structure Changes Investment Policy and Cash Flows



A	L
OA	D
	E
+ PV(DTS)	
- PV(CFD)	

- In a MM world, assume that firms pick OA that maximizes firm value (i.e., OA fell from sky)
  - ➢ May not be the case since Equity holders in control of the firm and pick projects, and the payoffs of Debt and Equity holders differ.
  - ➢ Equity payoff is convex while Debt payoff is concave
  - ➢ There is "conflict" in payoffs when cash flows are such that Debt cannot be paid in full (i.e., Debt is risky)
- <u>Three problems arise when debt is risky</u>
  - ➢ Wealth Transfer: A sophisticated way of saying "stealing". Equity holders take some projects that involve moving cash from debt holders into their own pockets
  - ➢ Risk Shifting: Equity holders take some risky negative NPV projects. Why? Because they know that in some states of the world the costs of the project are borne by Debt holders but they get all the benefits. Thus, equity holders "shift" the risk to Debt holders
  - ➢ Debt Overhang: Equity holders do not want to take some positive NPV projects. Why? Because some of the benefits of the project accrue to existing debt holders since this debt is "hanging" over their heads (not paid in full in some states)
- Broadly, these can also be thought of as CFD

Lecture 8A

Relaxing Assumption #4:

Capital Structure Changes Investment Policy and Cash Flows

Wealth Transfer

- Project Chariot
- Banks paying dividends from bailout money
- Other "stealing"

Risk Shifting

- S&L crisis
- Banks before subprime crisis
- AIG

Debt Overhang

- Project Chariot
- Greece and Euro
- Banks not lending (crisis)

Lecture 8A

Relaxing Assumption #4:

Capital Structure Changes Investment Policy and Cash Flows

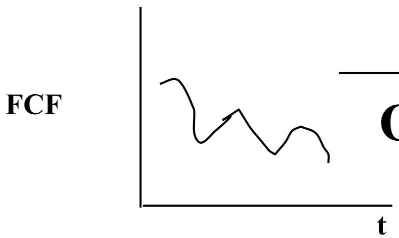
- Conflicts between managers and outside shareholders
  - Shirking
  - Quiet Life / Not putting forth enough effort
  - Building pet projects ("Free Cash Flow" Problem)

- Solution: Corporate Governance
  - Monitoring by boards
  - Compensation incentives
  - Debt or dividends
    - Jensen's solution to FCF problem

Lecture 8B

Relaxing Assumption #5:

Information is Asymmetric



A	L
OA	D
	E
+ PV(DTS)	
- PV(CFD)	

- In a MM world, info "symmetric" b/w firms and market
  - But firms may have private info, i.e. info is asymmetric
  - If info is value relevant, firms may want to communicate this info to market participants using "signals" (certain actions)
  - Capital structure and payout policy can be used as signals. Thus, capital structure may be relevant for firm value.
- How and when can such signals (actions) be used? TWO principles important for understanding this
  - **Credibility:** Signal from firms that have info (good firms) taken seriously by market (i.e., impacts firm value) if firms without info (bad firms) cannot copy the signal costlessly
  - **Adverse Selection**: In presence of asymmetric info, buyers may discount the value of a good since they think that sellers might be selling them "lemons"
  - ➔ Debt/dividends/repurchases/CEO buying shares are positive signals and selling equity is a negative signal
- Cost of external finance reduced for firm if information "insensitive" securities are used to finance projects
  - Gives "**Pecking Order" theory of capital structure**. For financing investments, use (1) Earnings, then (2) Debt, and last (3) Equity. This explains why profitable firms have low debt.

Lecture 8B

Relaxing Assumption #5:

Information is Asymmetric

Raising equity is costly due to adverse selection discount

Pinduoduo to Raise More Than $1 Billion in Alibaba Challenge

By Drew Singer and Meghan Genovese
February 5, 2019, 7:45 PM CST *Updated on February 6, 2019, 8:50 AM CST*

"The shares fell 4.6% in US trading Wednesday [Feb 6] to $28.94."

IntriCon down 5.5% on pricing 1.5M equity offering

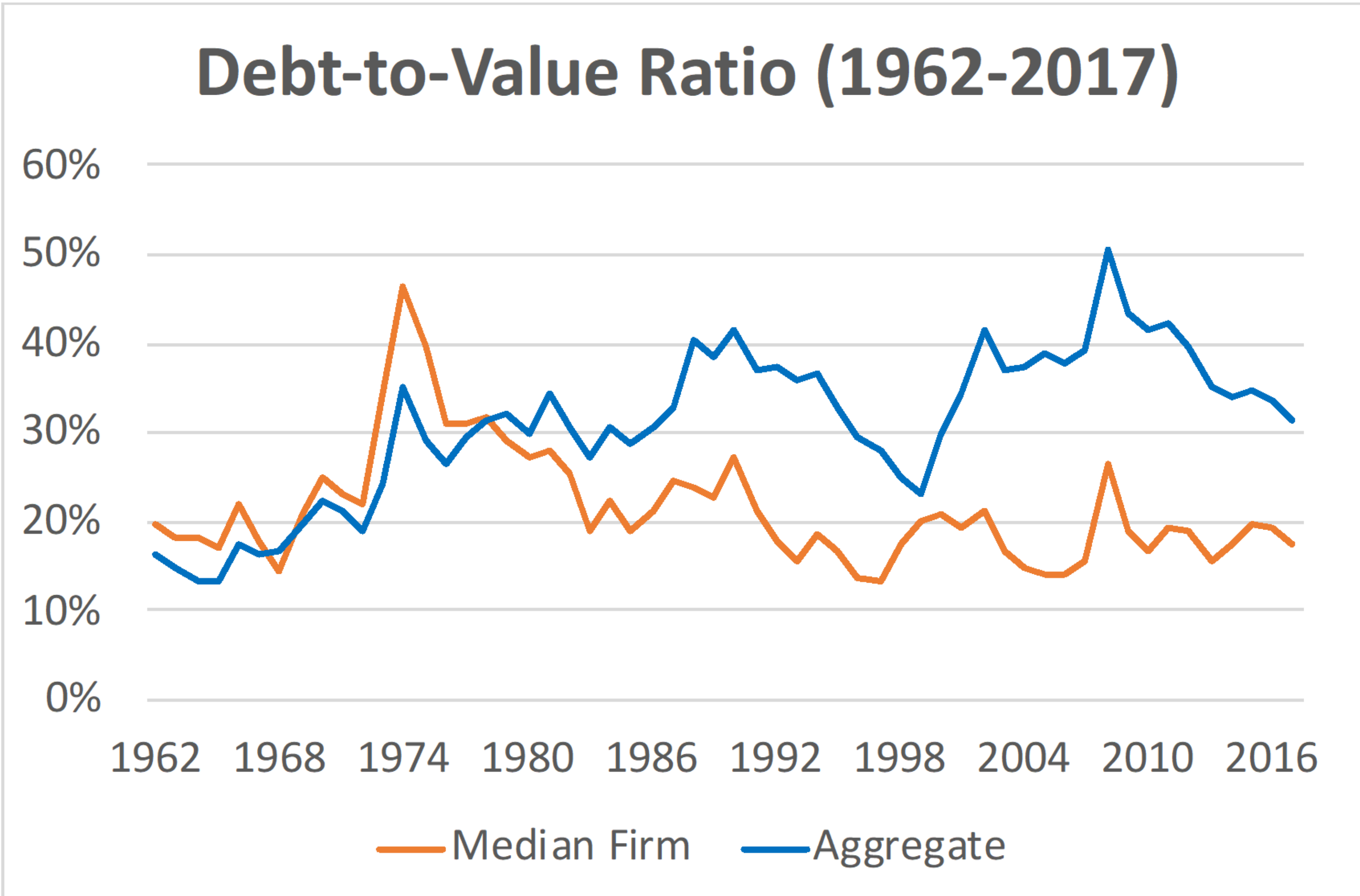
Aug. 16, 2018 6:09 AM ET | About: IntriCon Corporation (IIN) | By: Niloofer Shaikh, SA News Editor

Marriott Case

- Why change structure of the firm?
  - Benchmark against MM
    - Reduce debt overhang
  - Why do debt holders care?
- How can we compare it to the examples in lectures?
  - Seems like wealth transfer from D to E

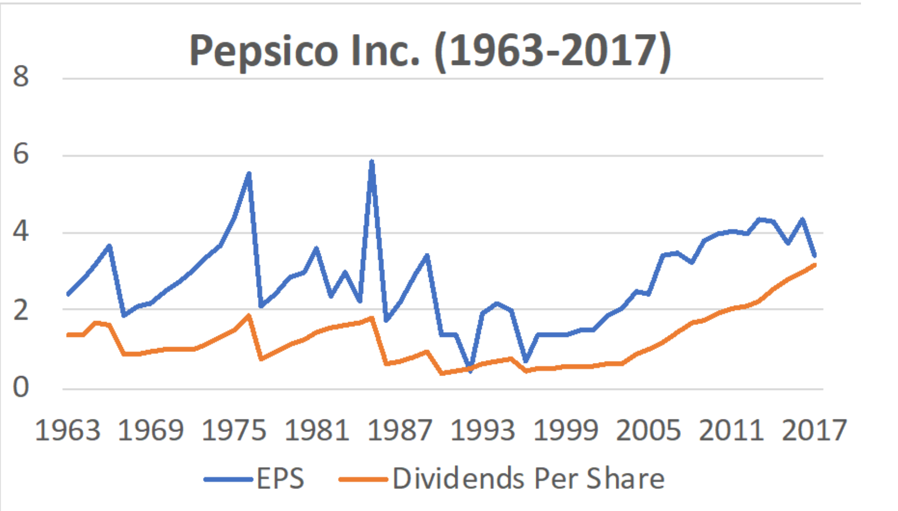
- Issues to tackle
  - Fiduciary duty of board
    - Toward shareholders?
    - Why should they automatically care about firm reputation?
  - Legal
    - "Event" risk covenants provide protection to D

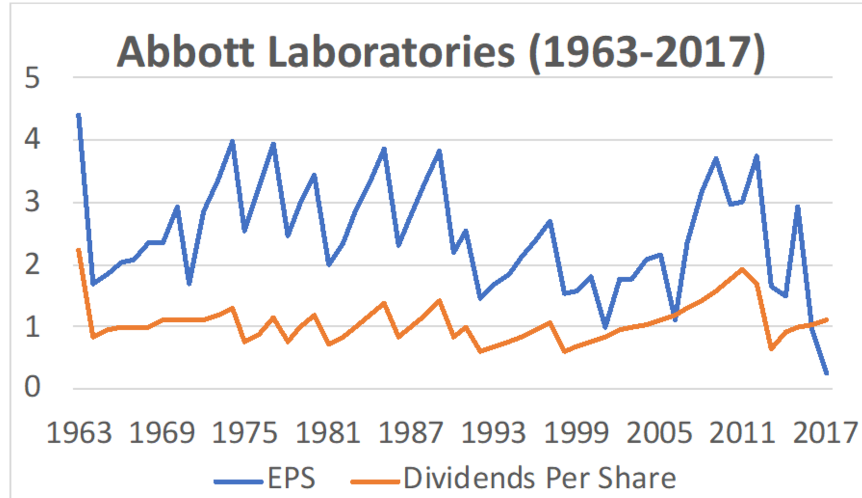
Real World: D/V

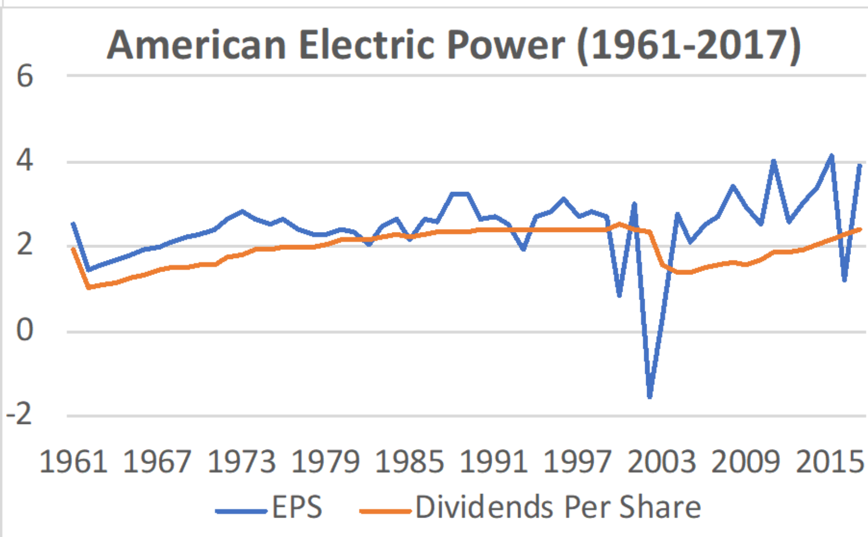


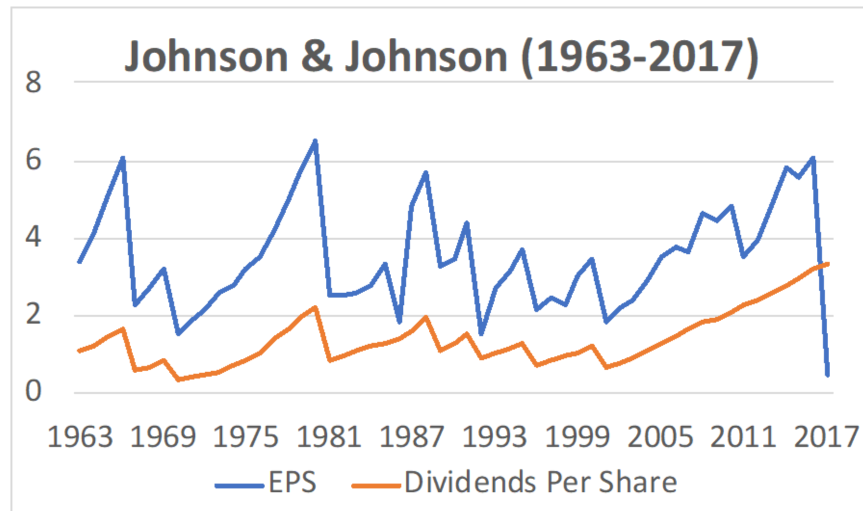
Data: Compustat

Real World: Dividends





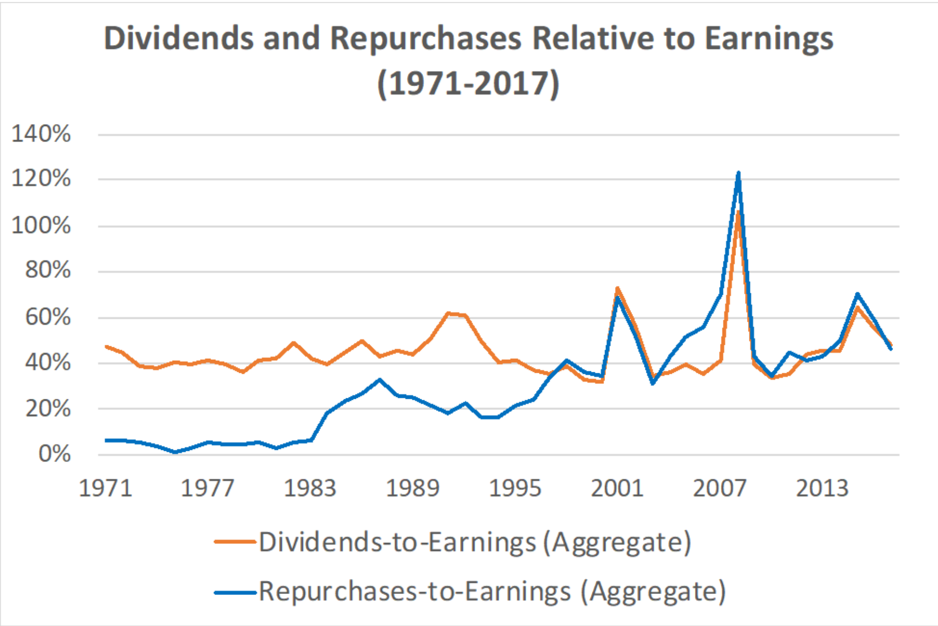


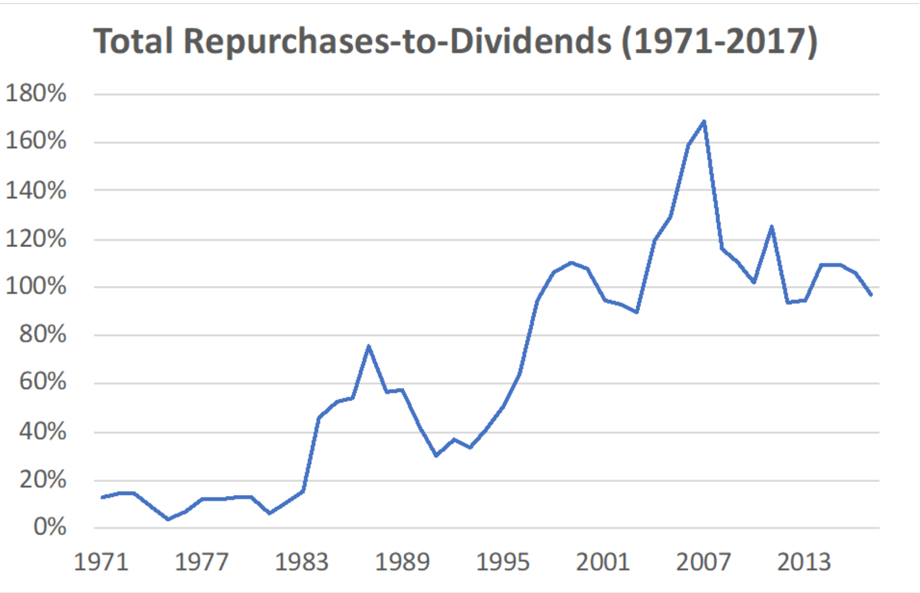


- **Dividends are sticky**
- **Once started, firms fight hard not to cut dividends**

Data: Compustat

Real World: Dividends vs. Repurchases





- **Progressively more $ paid to shareholders through repurchases**

Negatives of Dividends

- Higher taxes on investors
- CEO comp (options) negatively impacted
- A “long-term” commitment

Data: Compustat

Final Exam

Preparing for Final Exam

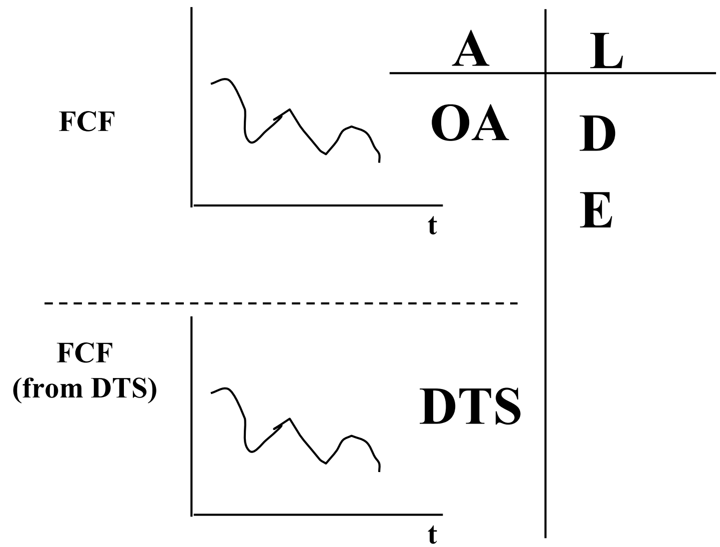
- Emphasize issues in
  - Lectures…and…
  - Case discussions (concepts, not numbers)
    - Similar to midterm
- Suggested strategy
  - Review lecture notes **&** case discussions
  - Take the practice final and review solutions
  - Do posted practice problems

What are the Takeaways?

Two Principles

1. How firms should make investment decisions
   - In an ideal world, take those with NPV>0

$$\boxed{V = \sum_t \frac{CF_t}{(1+r)^t}}$$



$$r_A = R_F + \beta_A(E[R_M - R_F])$$
$$FCF_t = EBIT_t(1-\tau_C) + Dep_t - CAPX_t - \Delta NWC_t$$

2. How firms create or destroy value with financial decisions (eg: capital structure decisions, payout decisions)
   - View deviations from MM world to figure this out

The End